L4. Experiments

Experiment

Experiments are a *special type* of observational process. What makes them special is that they **actively change the system** in order to learn about causal relationships. An experiment has three defining features: **Control of Background Variables (through manipulation)** , **Intervention on the Target Variable (through manipulation)** **and** **Observation of the Difference Produced by the Intervention.**

Control of Background Variables (through manipulation)

What is it?

A person might have an issue with a machine, before figuring out what caused the failure of the machine (a set of hypothesis) the person must **reduce or eliminate the influence of other factors** that could interfere with the result. The person **sets up conditions** so that only the variable of interest can make a difference.

In simple words:

Remove **anything** that can affect the thing you are trying to test.

Examples:

- Turning off all appliances before testing a single one in the fuse-box example.

- Keeping temperature, humidity, and soil constant when testing fertilizer on plants.
- Giving the control group a placebo so that belief in treatment does not bias results.

Intervention on the Target Variable (through manipulation)

What is it?

Once the background is controlled, the experimenter **changes the independent variable** — the factor believed to cause the effect. This is the **intervention**, and it is most important part. Without intervention, we only have an observational study, not an experiment.

In simple words:

I have made sure that everything that can affect the thing i'm trying to test is removed. Now, I test my hypothesis by changing the thing.

Example:

- Switching on the lamp to see if it blows the fuse.
- Giving a drug to the test group.
- Increasing voltage in an electrical circuit.

Observation of the Difference Produced by the Intervention

What is it?

Now the experimenter **compares the outcome before and after the intervention.** If the results show that there is a difference before and

after the intervention is made, we have a strong case to believe that our hypothesis H_0 is valid.

Example:

- The results show that the fuse blows only when the lamp is turned on → the lamp is the cause.
- The results show that plants grow taller only when fertilizer is used → fertilizer has a positive effect.
- The results show that there weren't any difference in speed after and before we installed new tires on the car → the tires do not improve speed.

Mill's Method of Difference

What is it?

Mill's Method of Difference is a strategy to figure out **what causes a certain effect**.

It goes like this : if you create **two situations that are identical in every way except one factor**, and only one of them shows the effect, then that differing factor is likely the cause.

In simple words:

> If everything is the same except one thing, and only one situation produces the outcome, then that one thing caused it.

- We **control** everything except the suspected cause
- We **intervene** on the suspected cause
- We **observe** whether the effect appears only when the cause is present

  If yes → strong causal evidence

  If no → hypothesis rejected or revised

Examples:

- **Question:** What caused the fuse to blow? (E)
- **Hypothesis:** The lamp caused the fuse to blow (C)
- **S1:** Lamp ON, all other appliances OFF → fuse blows
- **S2:** Lamp OFF, all other appliances OFF → fuse does *not* blow

Since the **only difference** between the two situations was the lamp being turned on, and the fuse blew only in S1, we infer that **the lamp caused the fuse to blow**.

Difference between Observation and Experiment

- **Observation**: We *only watch* what is already happening. No variable is actively changed.

  Goal: Detect patterns, not causes.

  Example: Measuring the temperature of lakes at different depths, or watching the orbits of Saturn's moons.

- **Experiment**: We *actively change* a variable (intervention) and observe the effect while controlling background factors.

  Goal: Test causal hypotheses ("Does X cause Y?").

  Example: Giving one group a new medicine and another a placebo, or changing the pressure in a chamber to see how it affects boiling point

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Observational Study

What it is:

A study where the researcher **does not intervene** in the system but only **observes what is already happening**.

Some background factors may still be *controlled*, but only by choosing when or how to observe not by manipulating the system itself.

In other words:

Just observing. No intervention → no active manipulation of the variable we want to test.

Example:

- Comparing wages of workers in dangerous vs. safe jobs without changing their jobs.
- Recording star movements through a telescope.
- Measuring the temperature of your hand.

Natural Experiment

What it is:

A natural experiment is **not a real experiment**, because we do not **not** change anything.

Instead, something in the real world **happens on its own** and creates two groups that are different in just one important way, almost as if we had set up an experiment.

So we **only observes** what already happened, but the situation is useful for studying cause and effect because the groups can be compared like in a real experiment.

Simpler words:

> Nature (or society) creates the "experiment," not the scientist, i.e we do not make any **intervention** on the variable. The world has already *provided* a situation where the background variables are the same (or similar) except for the one variable of interest.

Why it works:

The background conditions are already controlled by chance, luck, or external events so we can sometimes make causal claims even without doing any intervention.

Example:

- In 1854, John Snow studied a cholera outbreak in London.

  People got water from different wells. One well was contaminated, the others were clean.

  Snow did not assign people to water sources they already lived near them.

  Because the only major difference between the sick and healthy groups was the water source, he could conclude that **dirty water caused cholera**.

Another example:

- A new law is passed in only one city but not in another. Researchers can compare the two cities "as if" one was a treatment group and the other a control group.

Field Experiment

What it is:

A **field experiment** is an experiment done **in the real world** instead of in a laboratory, here we can change the variable of interest (intervention) but we can't successfully control all background variables because it might be difficult.

- We **do intervene** on the independent variable
- But we **cannot fully control** all background variables, because the environment is natural, not artificial.

Example:

- Researcher applies fertilizer to **some** farm plots (intervention)
- Other plots are left unchanged (control group)
- But weather, insects, soil nutrients, etc. **cannot be fully controlled, so** they can affect the results.

Laboratory Experiment

What it is:

A **real experiment** where the researcher not only manipulates the independent variable, but also **controls (or eliminates) nearly all background variables** . Compared to the other experiments like **field experiment** and **natural experiment (note that natural experiment is a observational study** where conditions mimic an experiment**)**, we have full control of the environment and can make interventions.

Example:

- Testing the boiling point of a liquid by changing pressure inside a controlled pressure chamber.
- Chemical reaction performed in a lab under fixed temperature and humidity.
- Medical trials done under controlled hospital/lab conditions with blinding and randomization.

Simulation Experiment

What is it?

A *simulation experiment* is a study where we **build a model (usually on a computer)** that behaves *like* a real experiment, but the things we manipulate are **not real physical variables** —they are **representations** inside the model.

So instead of changing real temperature, real pressure, or real patient behavior, we change **numbers, equations, or programmed conditions** that *stand in* for the real-world variables.

In simpler words:

We manipulate a model of reality, not reality itself.

Therefore, it is *not* a true experiment according to the book chapter 4.

Example:

A city planner simulates traffic flow and tests what happens if a highway exit is removed.

→ Cars in the model are not real cars so the change affects a representation.

Experimental Error

Observer Effect / Influence Problem

What it means:

The act of observing **changes the thing being observed**.

Example:

Putting a thermometer in a liquid slightly changes the liquid's temperature because it must absorb some thermal energy to take a reading .

Or: animals behave differently when they know a human is watching.

Confirmation bias

What is it?:

The results appear distorted because the **observer sees what they expect or want to see**, not what is actually there.

Example:

A researcher who strongly believes the drug works may *unconsciously read* slightly lower pain scores in the treatment group, even when scores are similar.

Placebo Effect

What it means:

A patient improves because they **believe they received treatment**, even if the pill contains no active ingredient.

Example:

A sugar pill reduces pain simply because the patient *expects* it to.

Selection bias

What it means:

The way a sample (or groups within a study) is chosen **is not random**, and ends up being influenced by a factor that affects the outcome.

Example:

Studying the effects of exercise only on people who *chose to join a gym* who are already healthier than average.

Internal Validity

What it means:

A study has **internal validity** when the effect we observe is *really caused* by the factor we are studying and not by some uncontrolled background variable.

> **In simpler words:**
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> "Did X really cause Y, or was something else responsible?" If this is true, then the change in the dependent variable can be confidently attributed to the intervention, not to chance, bias, or hidden influences.

Example:

A drug trial has high internal validity if the only difference between the treatment and control groups is the drug itself - not age, diet, or stress levels.

Detecting Experimental Errors

Repetition

What it means:

Doing **exactly the same study again**, following the same procedures — same design, same variables, same method.

Example:

Another lab performs the same drug trial using the same protocol and same type of patients.

Reproduction

What it means:

A repetition that **gets the same result** as the original study.

Example:

If the second trial confirms the same drug effectiveness, the result is *reproduced*.

Replication

What it means:

Doing a **modified version** of the study but with slightly different conditions, population, tools, etc. This is to check if the result still holds.

Example:

Testing the same drug but on older patients, or in a different country, or using a different measurement method.

Constancy

What it is:

Keeping background factors **the same** across all groups or trials so they cannot affect the result.

Example:

- Testing two fertilizers on plants but keeping soil type, sunlight, and watering *constant* in both groups.

Elimination

What it is:

Completely **removing** a background factor so it cannot have any influence at all.

Example:

- Dropping objects in a vacuum chamber to eliminate air resistance.
- Using a Faraday cage to eliminate radio interference during measurements.

Binding

What it is:

A strategy to prevent the **observer effect** or **confirmation bias** by **hiding information** from participants or experimenters so they cannot influence the results (consciously or unconsciously).

Single Blinding / Subject Blinding

What it is:

Only the **participants** do not know whether they are in the treatment or control group.

Example:

- In a drug test, patients get either a real pill or a placebo, but they do not know which one.

Experimenter Blinding

What it is:

The **researcher** collecting or interpreting data does **not know** which subjects are in the control or treatment group.

Example:

A lab technician measures response levels but has no information about who received the real drug.

Double Blinding

What it is:

Both **participants** and **experimenters** are blinded, neither knows who is in which group.

Example:

A medical trial where patients do not know if they received a real drug or placebo, and researchers taking measurements also do not know.

This is considered the **gold standard** in clinical experiments.

Separation

What it is

Separation is a strategy used in experiments when we **cannot eliminate** a background factor, but we can still **measure its effect independently** and remove or correct for it afterward.

So instead of:

- **Constancy** → keeping the background factor the same
- **Elimination** → removing the background factor entirely

Separation means:

> "We let the background factor remain, but we monitor it separately so we can mathematically subtract or control for its effect."

In other words, the background factor is **not removed**, but its influence is **isolated and accounted for**.

Example:

- If there is constant ambient sound in the factory, you measure the room noise first, then subtract it from the machine noise.
- If someone is nervous because of the lab environment, you first measure their baseline heart rate before exercise. Then you subtract the baseline from the exercise measurement.

Randomisation in Experiments

Randomization

What is it?

A method where a **random process** (coin flip, lottery, computer RNG) is used to assign units from a sample into **treatment and control groups**.

The goal is to prevent biased assignment and make groups comparable.

Examples:

- Flipping a coin to decide whether each participant gets bright light therapy or placebo.
- Using a random-number generator to divide 100 test samples into two groups.
- Assigning students to two teaching methods by shuffling names digitally.

Random Sampling

What is it?

A method where a **random process** is used to select the sample **from the population**.

This supports **external validity** because the sample is more likely to represent the population.

Examples:

- Drawing 500 names at random from a national database to study mental health.
- Using random-digit dialing to select survey participants.
- Randomly choosing 40 engines from a factory production line for testing.

Data Stratification

What is it?

A procedure where the sample is **divided into subgroups (strata)** based on **known background factors** *before* randomization, so that these factors are balanced across treatment and control groups.

Examples:

- Separating participants by age group (e.g., under 40 / over 40) and randomizing within each group.
- Stratifying patients by disease severity before assigning them to drug vs placebo.
- Grouping plants by initial height before randomizing fertilizer treatment.

External Validity

What is it?

External validity is about how well the results of a study **can be generalized beyond the study itself**.

In other words, it asks:

> “Do the findings hold only for the specific experimental setup, or also in the real world / in other populations / at other times / in other conditions?”

A study has high external validity when its conclusions apply **not just to the sample tested**, but also to the **larger target population or context we care about**.
